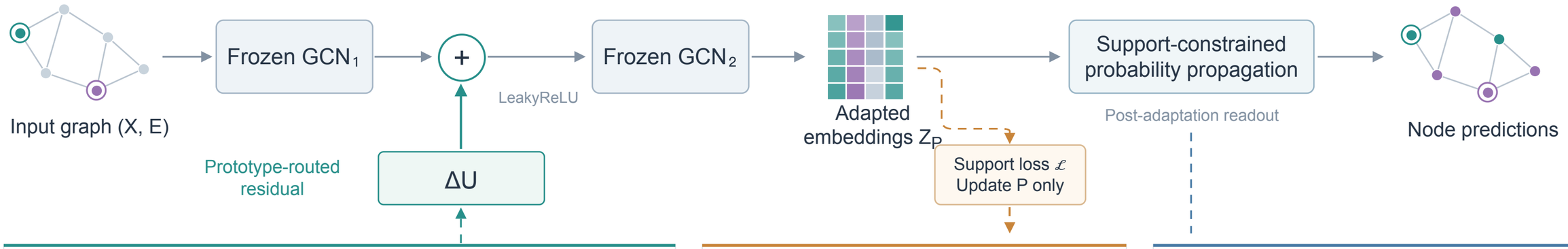
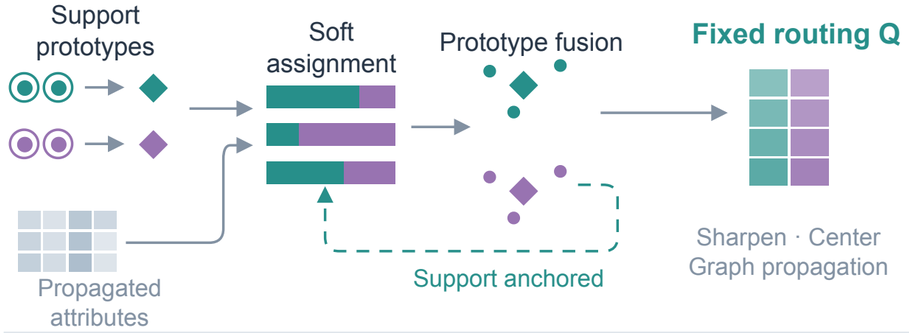
PRPrompt

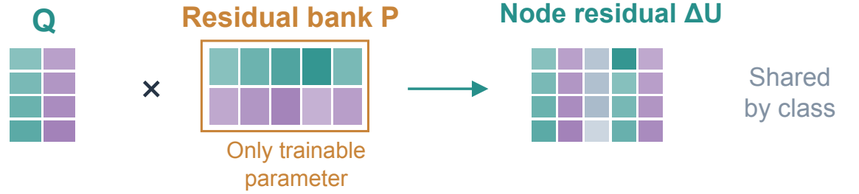
Prototype-routed graph prompting



M1 Prototype routing & shared residuals

Support classes define routing; P learns the residual

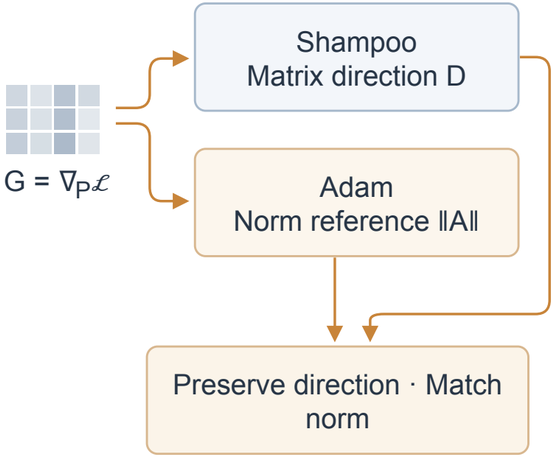




$$\Delta U = \gamma\,[Q \tanh(P)] \odot s_U$$

M2 Matrix update

One gradient; direction and step size

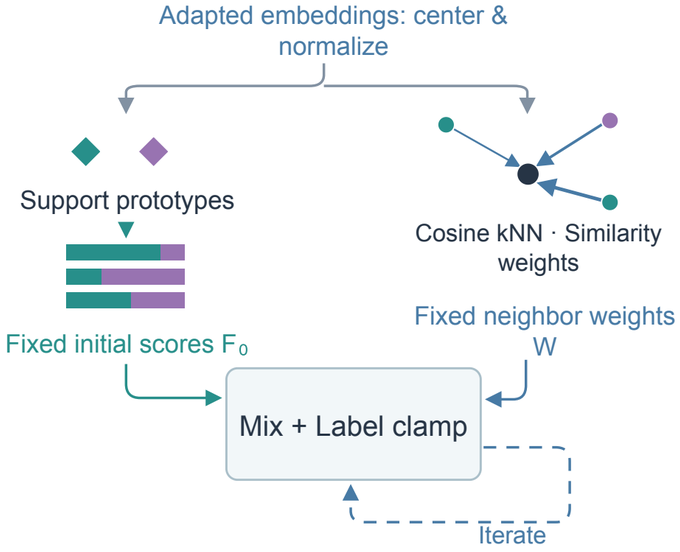


$$\tilde{D} = D \cdot \|A\|_F / (\|D\|_F + \varepsilon)$$

$$\mathbf{P \leftarrow P - \eta \tilde{D}}$$

M3 Support-constrained readout

Final embeddings fixed; iterate probabilities



$$F \leftarrow (1-\alpha)F_0 + \alpha W F \;;\; F_S \leftarrow Y_S$$